

Final Course Breakdown (100+ Lectures, 8-10 Hours)

Section 1: Introduction & Course Overview (30 min, 7 Lectures)

- 1.1 Welcome & Course Expectations (3 min)
- 1.2 What is an API Gateway? (5 min)
- 1.3 Why Use API Gateways? (5 min)
- 1.4 Introduction to Apigee & Its Key Features (5 min)
- 1.5 Setting Up Your Apigee Account (6 min)
- 1.6 Navigating the Apigee Interface (6 min)
- 1.7 Quick Demo: Creating Your First API Proxy (5 min)

Total: 7 Lectures

Section 2: API Management Fundamentals (70 min, 10 Lectures)

- 2.1 Introduction to API Management (4 min)
- 2.2 Why is API Management Important? (4 min)
- 2.3 Key Principles of API Management (4 min)
- 2.4 What Are API Proxies? (5 min)
- 2.5 API Gateway vs API Proxy (5 min)
- 2.6 Creating an API Proxy in Apigee (6 min)
- 2.7 Understanding Policies in Apigee (6 min)
- 2.8 Configuring Basic Policies (6 min)
- 2.9 Troubleshooting API Proxies (7 min)
- 2.10 Debugging API Issues in Apigee (7 min)

Total: 10 Lectures

Section 3: RESTful APIs & OpenAPI Integration (75 min, 12 Lectures)

- 3.1 What are RESTful APIs? (4 min)
- 3.2 Key REST API Methods (4 min)
- 3.3 Understanding HTTP Status Codes (4 min)
- 3.4 OpenAPI Specifications Overview (5 min)
- 3.5 Benefits of OpenAPI in API Development (5 min)
- 3.6 Importing OpenAPI Specs into Apigee (6 min)
- 3.7 Configuring API Proxies for OpenAPI (6 min)
- 3.8 Understanding JSON and XML Payloads (5 min)
- 3.9 Handling JSON Data in APIs (5 min)
- 3.10 Handling XML Data in APIs (5 min)
- 3.11 API Testing with Postman (6 min)
- 3.12 API Testing with JMeter (6 min)

Total: 12 Lectures

Section 4: API Security & Authentication (90 min, 14 Lectures)

- 4.1 Introduction to API Security (5 min)
- 4.2 Common API Security Risks (5 min)
- 4.3 What are API Keys? (5 min)
- 4.4 Implementing API Keys in Apigee (6 min)
- 4.5 Introduction to OAuth 2.0 (5 min)
- 4.6 OAuth 2.0 Grant Types Explained (5 min)
- 4.7 Implementing OAuth 2.0 in Apigee (6 min)
- 4.8 What is JWT Authentication? (5 min)
- 4.9 Implementing JWT Authentication in Apigee (6 min)
- 4.10 Securing APIs with Mutual TLS (5 min)
- 4.11 Implementing Mutual TLS in Apigee (6 min)
- 4.12 Rate Limiting in API Security (5 min)
- 4.13 API Threat Management Best Practices (6 min)
- 4.14 Securing Generative AI APIs (6 min)

Total: 14 Lectures

Section 5: Managing Generative AI APIs (90 min, 15 Lectures)

- 5.1 Introduction to AI API Management (5 min)
- 5.2 Unique Challenges of AI API Management (5 min)
- 5.3 Setting Up an AI API Proxy in Apigee (6 min)
- 5.4 Configuring API Proxies for AI APIs (6 min)
- 5.5 API Mashups: Combining AI APIs (6 min)
- 5.6 Caching AI API Responses (6 min)
- 5.7 Rate Limiting for AI APIs (5 min)
- 5.8 Traffic Management for AI APIs (6 min)
- 5.9 Optimizing AI API Performance (6 min)
- 5.10 Monitoring AI APIs in Apigee (6 min)
- 5.11 Logging and Debugging AI APIs (6 min)
- 5.12 Scaling AI APIs Efficiently (6 min)
- 5.13 Cost Management Strategies for AI APIs (6 min)
- 5.14 Security Considerations for AI APIs (6 min)
- 5.15 Real-World AI API Case Study (6 min)

Total: 15 Lectures

Section 6: API Lifecycle & Developer Portal (75 min, 12 Lectures)

- 6.1 Understanding API Lifecycle (5 min)
- 6.2 API Versioning Best Practices (5 min)
- 6.3 Creating an API in Apigee (6 min)

- 6.4 API Publishing in Apigee (6 min)
- 6.5 Managing API Deprecation (6 min)
- 6.6 Introduction to Developer Portals (5 min)
- 6.7 Setting Up a Developer Portal (6 min)
- 6.8 Customizing Developer Portals (6 min)
- 6.9 Managing API Documentation (6 min)
- 6.10 Real-Time Monitoring and Analytics (6 min)
- 6.11 API Monetization Strategies (6 min)
- 6.12 Managing API Consumers (6 min)

Total: 12 Lectures

Section 7: API Performance Optimization & Monitoring (75 min, 12 Lectures)

- 7.1 Introduction to API Performance Optimization (5 min)
- 7.2 Identifying API Performance Issues (6 min)
- 7.3 Caching Strategies for API Performance (6 min)
- 7.4 Throttling API Traffic (6 min)
- 7.5 Optimizing API Response Times (6 min)
- 7.6 Real-Time API Monitoring in Apigee (6 min)
- 7.7 Setting Up API Logs in Apigee (6 min)
- 7.8 Monitoring AI API Workloads (6 min)
- 7.9 API Traffic Analytics & Reporting (6 min)
- 7.10 Scaling APIs for High Traffic (6 min)
- 7.11 Handling API Failures & Downtime (6 min)
- 7.12 Advanced Debugging in Apigee (6 min)

Total: 12 Lectures

Section 8: DevOps & CI/CD for API Deployment (75 min, 11 Lectures)

- 8.1 Introduction to CI/CD in API Management (5 min)
- 8.2 What is API Versioning in CI/CD? (6 min)
- 8.3 Versioning APIs with Git (6 min)
- 8.4 Automating API Testing (6 min)
- 8.5 Setting Up a CI/CD Pipeline (6 min)
- 8.6 Rolling Back API Changes (6 min)
- 8.7 Deploying APIs with Zero Downtime (6 min)
- 8.8 Securing CI/CD Pipelines (6 min)
- 8.9 Monitoring API Health During CI/CD (6 min)
- 8.10 Troubleshooting CI/CD Pipeline Issues (6 min)
- 8.11 Best Practices for API CI/CD Workflows (6 min)

Total: 11 Lectures

Section 9: Real-World Case Studies & Best Practices (60 min, 10 Lectures)

- 9.1 Real-World API Gateway Case Study (6 min)
- 9.2 Handling Large-Scale AI APIs (6 min)
- 9.3 Best Practices in API Security (6 min)
- 9.4 API Design Patterns & Antipatterns (6 min)
- 9.5 Understanding API Business Models (6 min)
- 9.6 API Monetization Strategies (6 min)
- 9.7 Common API Management Mistakes (6 min)
- 9.8 Post-Course Resources & Learning Path (6 min)
- 9.9 Course Wrap-Up & Next Steps (6 min)
- 9.10 Bonus: API Job Market Insights (6 min)

Total: 10 Lectures